The gradient of f is

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = c,$$

for any $x \in \mathbb{R}^n$. As it is a constant vector, the second derivative matrix is zero, for any $x \in \mathbb{R}^n$:

$$\nabla^2 f(x) = 0.$$

We first note that the value of b is irrelevant to determine the necessary optimality conditions. Adding a constant to an objective function does not change the optima. If $c = 0$, that is, if

$$c_1 = \ldots = c_n = 0,$$

then the necessary optimality conditions are verified for all $x \in \mathbb{R}^n$. If $c \neq 0$, that is, if there is at least one i such that $c_i \neq 0$, then the first order necessary optimality conditions are not verified . The second order necessary conditions are always verified, as the null matrix is positive semidefinite.

Note that, if $c = 0$, the linear function is constant, with value b. And any $x \in \mathbb{R}^n$ is a (global) optimum. If not, the function is not constant. As there is no constraint, it is not bounded and there is no optimum.